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Exploratory Data Analytics

(E1CSA231)

Micro Project Report

On

Spotify Data Analysis

Title of the Micro Project

Submitted by

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Abstract

- This project performs exploratory data analysis on a music dataset to understand song characteristics and popularity trends. Python libraries such as Pandas, Matplotlib, and Seaborn are used for data processing and visualization. The analysis examines the distribution of explicit and non-explicit songs and identifies the top music genres based on average popularity. It further explores relationships between features like danceability, tempo, and popularity using scatter plots. Histograms and box plots are used to study the distribution and spread of key variables. A correlation heatmap is generated to highlight relationships among different audio features. Overall, the study provides meaningful insights into the factors influencing song popularity and listener preferences.

- **Dataset Name:** Spotify Dataset

- **Source:** Kaggle.com

- **Purpose of This Analysis:** The purpose of this analysis is to examine how different song features influence popularity.It aims to identify patterns in genres, tempo, and danceability using visualizations.The study also explores the relationship between explicit content and song success. Overall, it helps in understanding trends and listener preferences in music data.

- **Tools Used:** Python, Pandas, Matplotlib, Seaborn

- **Key Finding:**

- Popularity varies significantly across genres, with a few genres dominating the top ranks.
- Danceability and tempo show some relationship with popularity, but not a very strong correlation.
- Explicit songs do not always have higher popularity compared to non-explicit songs.
- Correlation analysis indicates that no single feature strongly determines song popularity; it depends on a combination of factors

1.Introduction

1.1 Project Overview:

This project involves a comprehensive exploratory data analysis (EDA) of a music dataset to gain deeper insights into the factors influencing song popularity. The analysis is carried out using Python, leveraging powerful libraries such as Pandas for data preprocessing and manipulation, Matplotlib for basic plotting, and Seaborn for advanced statistical visualizations. The dataset contains various musical attributes including popularity, genre, danceability, tempo, loudness, and explicit content, which are systematically examined.

The project begins with data loading and cleaning, followed by univariate and bivariate analysis to understand individual features and their relationships. Visualizations such as pie charts are used to analyze the proportion of explicit versus non-explicit songs, while bar charts highlight the most popular genres. Scatter plots help in identifying relationships between numerical variables like danceability, tempo, and popularity. Histograms and box plots are used to study distributions and detect outliers.

Additionally, trend analysis is performed by sorting and plotting tempo and popularity values across the dataset. A correlation heatmap is generated to identify the strength and direction of relationships between multiple audio features. Through these methods, the project uncovers key patterns and dependencies within the data. Overall, the project aims to provide meaningful insights into listener preferences, support data-driven decision-making, and enhance understanding of music trends in the industry.

Furthermore, this project demonstrates the practical application of data analysis techniques in real-world scenarios, particularly in the music industry. By combining statistical analysis with visual storytelling, it enables better interpretation of complex data and supports informed decision-making. The insights gained can be useful for artists, producers, and streaming platforms to understand audience preferences, optimize content creation, and improve recommendation systems. This analysis also builds a strong foundation for future work, such as predictive modeling and machine learning applications

1.2 Objectives of the project:

- To perform exploratory data analysis (EDA) on the music dataset.
- To understand the distribution of different song attributes.
- To analyze the impact of explicit content on song popularity.
- To identify the most popular music genres.
- To examine the relationship between danceability and popularity.
- To study how tempo affects song popularity.
- To visualize data using various plots and charts.
- To detect patterns, trends, and outliers in the dataset.
- To analyze correlations among multiple audio features.
- To gain insights into listener behavior and music industry trends.

1.3 Problem Statement:

In the rapidly growing music industry, understanding what makes a song popular is a challenging task due to the presence of multiple influencing factors such as genre, tempo, danceability, and explicit content. There is a need to analyze and interpret these features to identify patterns and relationships that contribute to a song's success. This project aims to perform exploratory data analysis on a music dataset to uncover key insights, visualize trends, and determine how different audio attributes impact song popularity, thereby helping in better decision-making and understanding listener preferences.

Key Factors Affecting the Customers

- Genre – Certain music genres tend to attract more listeners and higher popularity.
- Danceability – Songs that are easier to dance to often gain more attention.
- Tempo – The speed of a song can influence listener preference and engagement.
- Explicit Content – Presence of explicit lyrics may impact audience reach.
- Loudness – Higher or balanced loudness can make a song more appealing.

- Energy – Energetic songs are often more engaging and popular.
- Acousticness – The level of acoustic elements can affect listener interest.
- Speechiness – Songs with more spoken words may appeal to specific audiences.
- Instrumentalness – Instrumental tracks may have different popularity trends.
- Liveness – Indicates if the song feels like a live performance, affecting appeal.

1.4 Scope of the Project:

This project includes the analysis of a music dataset to explore relationships between various audio features and song popularity. It covers data loading, cleaning, and exploratory data analysis using Python libraries such as Pandas, Matplotlib, and Seaborn. The project includes visualizations like pie charts, bar graphs, scatter plots, histograms, box plots, and correlation heatmaps to identify patterns, trends, and relationships among features such as genre, danceability, tempo, loudness, and explicit content. The study aims to provide meaningful insights into listener preferences and general music trends based on the available data.

However, the project does not include advanced techniques such as machine learning or predictive modeling for forecasting song popularity. It is limited to the given dataset and does not consider external factors like artist reputation, marketing strategies, social media influence, or cultural trends. Real-time data analysis and large-scale industry-level predictions are also not part of this study. Therefore, the findings are based only on the dataset used and may not fully represent the entire music industry.

Included in the Project

- Data loading and preprocessing of the music dataset
- Exploratory Data Analysis (EDA) using Python
- Visualization using pie charts, bar graphs, scatter plots, histograms, box plots, and heatmaps

- Analysis of features like popularity, genre, danceability, tempo, and explicit content
- Identification of patterns, trends, and relationships between variables
- Correlation analysis of multiple audio features
- Interpretation of results to understand music trends and listener preferences

Not Included in the Project

- Machine learning or predictive modeling techniques
- Real-time or live data analysis
- External factors such as artist fame, marketing, or social media influence
- Large-scale industry forecasting or recommendation systems
- Data collection (dataset is pre-provided)
- Advanced statistical modeling or deep analysis beyond EDA
- Consideration of cultural or regional music preferences

1.2 Data Description

2.1 Source of data:

The dataset used in this project is obtained from a CSV file named **"spotify dataset.csv"**, which was sourced from **Kaggle.com**. The dataset contains information about songs, including features such as popularity, genre, danceability, tempo, and other audio characteristics. It is used for performing exploratory data analysis to identify patterns, relationships, and trends in music data.

2.2 Data Overview:

The dataset used in this project is the **Spotify dataset**, which contains detailed information about various songs and their audio features. Each row in the dataset represents a song, while the columns describe different attributes such as popularity, track genre, danceability, tempo, loudness, speechiness, acousticness, instrumentalness, liveness, duration, and whether the song is explicit or not.

The dataset includes both numerical and categorical variables, making it suitable for comprehensive exploratory data analysis. Numerical features help in analyzing relationships and distributions, while categorical features like genre and explicit content help in comparing groups. Overall, the dataset provides a structured view of music characteristics, enabling the identification of trends and factors influencing song popularity.

2.3 Features:

Column Name	**Description**
Track Name	Name of the song
Artist Name	Name of the artist
Track Genre	Genre/category of the song
Popularity	Popularity score of the song (0–100)
Duration_ms	Length of the song in milliseconds
Explicit	Indicates if the song has explicit content (True/False)

Column Name	Description
Danceability	How suitable the song is for dancing
Energy	Intensity and activity level of the song
Loudness	Overall loudness (in decibels)
Speechiness	Amount of spoken words in the song
Acousticness	Likelihood of the song being acoustic
Instrumentalness	Probability that the song has no vocals
Liveness	Likelihood of the song being performed live
Tempo	Speed of the song (beats per minute - BPM)

3.Data Preprocessing

3.1 Data Cleaning:

Data cleaning is performed to ensure the dataset is accurate, consistent, and free from errors. In this project, the dataset is first checked for missing (null) values, and rows with incomplete data are handled by either removing them or keeping only relevant entries. Duplicate records, if present, are identified and removed to avoid bias in the analysis.

Additionally, column names are reviewed and formatted properly, and data types are corrected where necessary (for example, ensuring numerical columns like popularity and tempo are in the correct format). Any irrelevant or unnecessary columns are excluded from the analysis. These steps help improve the quality of the dataset and ensure reliable results during visualization and analysis.

3.2 Handling missing values:

Handling missing values is an important step to ensure accurate analysis. In this project, the dataset is first checked for null or missing values using functions like is null() and sum (). Based on the importance of the features, missing values are handled appropriately to maintain data quality.

Rows with missing values in important columns such as popularity or tempo are removed to avoid incorrect analysis. In some cases, missing values can also be filled using suitable methods like mean, median, or mode, depending on the data type. By properly handling missing values, the dataset becomes more reliable and suitable for visualization and further analysis.

3.3 Data Transformation:

Data transformation involves preparing the dataset for analysis by selecting relevant features, correcting data types, and organizing the data properly. Categorical variables are structured for comparison, and data is sorted when needed to support visualization and better interpretation.

3.4 Outliers Detection:

Outliers are extreme values that differ significantly from other data points. They are identified using visualization methods like box plots. In this project, features such as popularity and tempo are checked for outliers. Detecting outliers helps improve the accuracy and reliability of the analysis.

4. Exploratory Data Analysis

4.1 Univariate Analysis

- Pie chart is used to show the proportion of explicit and non-explicit songs.
- Histograms are used to analyze the distribution of popularity values.
- Histograms are also used to study the distribution of tempo.
- Box plots help in identifying the spread of popularity values.
- Box plots are used to detect outliers in tempo.
- Helps in understanding the central tendency of individual features.
- Identifies skewness and variation in the data.
- Provides a clear overview of each feature independently.

Code:

```
# Pie Chart - Explicit vs Non Explicit
plt.figure()
df["explicit"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Distribution of Explicit and Non Explicit Songs")
plt.ylabel("")
plt.show()
# Histogram - Popularity
plt.figure()
df["popularity"].hist()
plt.title("Distribution of Song Popularity")
plt.xlabel("Popularity")
plt.ylabel("Number of Songs")
plt.show()
# Histogram - Tempo
plt.figure()
df["tempo"].hist()
```

```
plt.title("Distribution of Song Tempo")
plt.xlabel("Tempo")
plt.ylabel("Number of Songs")
plt.show()
# Box Plot - Popularity
plt.figure()
sns.boxplot(x=df["popularity"])
plt.title("Box Plot Showing Spread of Song Popularity")
plt.xlabel("Popularity")
plt.show()
# Box Plot - Tempo
plt.figure()
sns.boxplot(x=df["tempo"])
plt.title("Box Plot Showing Spread of Song Tempo")
plt.xlabel("Tempo")
plt.show()
```

4.2 Bivariate Analysis:

Bivariate analysis studies the relationship between two variables to understand how one affects the other. In this project, it is used to explore how song features influence popularity.

Code:

```
import matplotlib.pyplot as plt
import seaborn as sns
# Scatter Plot: Danceability vs Popularity
plt.figure()
plt.scatter(df["danceability"], df["popularity"])
```

```
plt.title("Relationship Between Danceability and Song Popularity")
plt.xlabel("Danceability")
plt.ylabel("Popularity")
plt.show()
# Scatter Plot: Tempo vs Popularity
plt.figure()
plt.scatter(df["tempo"], df["popularity"])
plt.title("Relationship Between Tempo and Song Popularity")
plt.xlabel("Tempo")
plt.ylabel("Popularity")
plt.show()
# Bar Plot: Explicit vs Average Popularity
plt.figure()
sns.barplot(x="explicit", y="popularity", data=df)
plt.title("Average Popularity of Explicit vs Non Explicit Songs")
plt.xlabel("Explicit")
plt.ylabel("Average Popularity")
plt.show()
```

4.3 Multivariate Analysis:

- Multivariate analysis studies relationships between multiple features at once.
- Correlation heatmap is used for features like popularity, tempo, danceability, and loudness.
- Shows which features are positively or negatively correlated with each other.

- Helps identify patterns not visible in single-variable or two-variable analysis.

CODE:

```
# Multivariate Analysis: Correlation Heatmap
plt.figure()
corr = df[["popularity","duration_ms","danceability","loudness",

"speechiness","acousticness","instrumentalness","liveness","tem
po"]].corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap of Music Features")
plt.show()
```

4.4 Data Visualization

- Pie chart shows explicit vs non-explicit songs.
- Bar charts compare popularity by genre and explicit content.
- Scatter plots show relationships between features like danceability and tempo with popularity.
- Histograms and box plots show distribution, spread, and outliers.
- Heatmap displays correlations between multiple numerical features.\

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load dataset
df = pd.read_excel("/content/Book 1.xlsx")
```

```
# Pie Chart
plt.figure()
df["explicit"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Distribution of Explicit and Non Explicit Songs")
plt.ylabel("")
plt.show()
# Bar Charts
# Average Popularity by Genre
plt.figure()
df.groupby("track_genre")["popularity"].mean().sort_values(ascending=False).head(10).plot(kind="bar")
plt.title("Average Popularity of Top 10 Music Genres")
plt.xlabel("Genre")
plt.ylabel("Average Popularity")
plt.show()
# Average Popularity of Explicit vs Non Explicit Songs
plt.figure()
sns.barplot(x="explicit", y="popularity", data=df)
plt.title("Average Popularity of Explicit vs Non Explicit Songs")
plt.xlabel("Explicit")
plt.ylabel("Average Popularity")
plt.show()

# Scatter Plots
# Danceability vs Popularity
plt.figure()
```

```
plt.scatter(df["danceability"], df["popularity"])
plt.title("Relationship Between Danceability and Song Popularity")
plt.xlabel("Danceability")
plt.ylabel("Popularity")
plt.show()
# Tempo vs Popularity
plt.figure()
plt.scatter(df["tempo"], df["popularity"])
plt.title("Relationship Between Tempo and Song Popularity")
plt.xlabel("Tempo")
plt.ylabel("Popularity")
plt.show()
# Histograms
plt.figure()
df["popularity"].hist()
plt.title("Distribution of Song Popularity")
plt.xlabel("Popularity")
plt.ylabel("Number of Songs")
plt.show()
plt.figure()
df["tempo"].hist()
plt.title("Distribution of Song Tempo")
plt.xlabel("Tempo")
plt.ylabel("Number of Songs")
plt.show()
# Box Plots
```

```
plt.figure()
sns.boxplot(x=df["popularity"])
plt.title("Box Plot Showing Spread of Song Popularity")
plt.xlabel("Popularity")
plt.show()
plt.figure()
sns.boxplot(x=df["tempo"])
plt.title("Box Plot Showing Spread of Song Tempo")
plt.xlabel("Tempo")
plt.show()
# Correlation Heatmap
plt.figure()
corr = df[["popularity","duration_ms","danceability","loudness",

"speechiness","acousticness","instrumentalness","liveness","tempo"]].corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap of Music Features")
plt.show()
```

4.5 Correlation Analysis

Correlation analysis studies the relationship between numerical variables to see how they influence each other. In your project, it is used to examine how features like popularity, danceability, tempo, loudness, speechiness, acousticness, instrumentalness, liveness, and duration relate to each other.

Code:

```
plt.figure()
corr = df[["popularity","duration_ms","danceability","loudness",
```

```
"speechiness","acousticness","instrumentalness","liveness","tempo"]].corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap of Music Features")
plt.show()
```

5.Results and Insights

➢ Key findings:

- Most songs in the dataset are **non-explicit**, but explicit songs do not always have higher popularity.
- **Popularity varies significantly across genres**, with some genres consistently more popular than others.
- **Danceability and tempo** show some relationship with popularity, but the correlation is weak to moderate.
- Distribution plots show that **most songs have moderate popularity and tempo**, with some outliers at extreme values.
- Box plots highlight **outliers** in both popularity and tempo, indicating variability in the dataset.
- The **correlation heatmap** shows that no single feature strongly determines popularity; it depends on a combination of factors like energy, danceability, and loudness.
- Overall, the analysis reveals **patterns and trends** that can help understand listener preferences and music characteristics influencing popularity.

➢ Patterns discovered:

- Most songs fall under **moderate popularity levels**, with fewer songs at extreme high or low values.
- Certain **genres consistently show higher average popularity** compared to others.
- **Danceability and tempo show slight positive trends** with popularity but are not strong predictors.
- The dataset contains **outliers in popularity and tempo**, indicating variation in song characteristics.
- **Correlation analysis shows weak to moderate relationships** among features, suggesting popularity depends on multiple factors.

➢ Business / Real-Life Meaning:

- Music companies can use this analysis to **identify popular genres and trends**.
- Artists can create songs with features that **match listener preferences**.
- Streaming platforms can improve **song recommendations and playlists**.
- Marketers can design better **promotion strategies** based on popular features.
- Helps in making **data-driven decisions** to increase song success.

➢ Simple Real-Life Insight:

- People generally prefer songs with **balanced tempo and good danceability**.
- **Genre plays an important role** in how popular a song becomes.
- **Explicit content does not guarantee higher popularity**.
- No single feature makes a song popular; it depends on a **combination of factors**.

6.Conclusion

This project focused on performing exploratory data analysis on a music dataset to understand the key factors that influence song popularity. The dataset was carefully processed, cleaned, and analyzed using Python libraries to ensure accurate and meaningful results. Through systematic analysis, important patterns and relationships within the data were identified.

A wide range of visualization techniques such as pie charts, bar charts, scatter plots, histograms, box plots, and heatmaps were used to explore the dataset. These visualizations helped in understanding the distribution of features, comparing different categories, and identifying relationships between variables like danceability, tempo, genre, and popularity.

The analysis showed that song popularity is not dependent on a single feature but rather on a combination of multiple attributes. While factors like genre, danceability, and tempo have some influence, their impact is not very strong individually. The presence of outliers and variations in the dataset also indicates that listener preferences can vary widely.

Correlation analysis further revealed that most features have weak to moderate relationships with each other, suggesting that predicting popularity is complex. This highlights the importance of considering multiple features together rather than relying on one single factor.

Overall, this project provides valuable insights into music trends and user preferences. The findings can be useful for artists, producers, and streaming platforms to make informed decisions regarding content creation, promotion, and recommendation systems. It also lays a foundation for future work involving advanced analytics and predictive modeling to better understand and forecast song popularity.

7.Future Scope

- Machine learning models can be applied to predict song popularity based on multiple audio features and improve accuracy of analysis.
- The study can be extended by using larger and real-time datasets to get more updated and reliable insights.
- Additional factors such as artist popularity, number of followers, and album success can be included for deeper analysis.
- Social media trends and user engagement data can be analysed to understand their impact on song popularity.
- Advanced recommendation systems can be developed to suggest songs based on user preferences and listening behaviour.
- A detailed genre-wise analysis can be performed to better understand audience interests in different types of music.
- Interactive dashboards and advanced visualization tools can be used to present insights more effectively.
- Deep learning techniques can be applied to capture complex patterns in music data.
- Regional and cultural preferences can be studied to understand variations in music taste across different locations.
- Integration with streaming platform data can provide real-time insights and help in making practical business decisions.

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